# SDMTS Supply Chain Mapping Methods Note

Data-cleaning, validation, dashboard, and mapping workflow

## Purpose

This methods note documents the analytical workflow represented in the public portfolio package. It is meant to show the logic of the project while avoiding publication of raw third-party company records or client working files.

## Data Workflow

- Import and inspect firm-level manufacturing records.
- Normalize company names, city fields, NAICS/SIC fields, and contact fields.
- Remove duplicate or obviously unusable records.
- Construct validation signals from phone, URL, D-U-N-S, address/geocode, and active-status indicators.
- Create a confidence score to prioritize likely-active businesses and records needing review.
- Generate latitude/longitude coordinates for map-ready records and manually review implausible coordinates.
- Group industries using three-digit NAICS codes to balance analytical detail and dashboard usability.
- Push cleaned records into dashboard input tables, pivot tables, mapping worksheets, and visual outputs.

## Dashboard Logic

- Slicers and filter tables allow users to narrow the dataset by categories such as city, region, and industry.
- Pivot tables and dashboard input sheets support charts such as supplier counts by city and industry.
- Search and lookup logic allows a user to retrieve information about a selected company record.
- A map update workflow connects filtered dashboard data to a map-ready output sheet.
- Protected and hidden sheets reduce the risk of accidental user changes to formulas or underlying data.

## Public Portfolio Treatment

The public workbook is intentionally redacted and uses synthetic company records. This avoids distributing proprietary D&B; Hoovers records, client working files, or firm-level data that was not intended as a public dataset. The workbook demonstrates the workflow structure, formulas, and dashboard logic rather than reproducing the original data.

## Limitations

- Confidence scoring is not a substitute for direct verification with a company or official public records.
- Web-search and automated signals can produce false positives or miss active businesses.
- Geocoding can produce incorrect coordinates, especially for ambiguous addresses.
- Map-based gap analysis should be treated as a starting point for follow-up due diligence.